

Priyanka Anil Lakariya

+1 617-516-4114 | Boston, MA | [LinkedIn](#) | lakariya.p@northeastern.edu | [Portfolio](#) | [GitHub](#)

EDUCATION

Northeastern University | *Master of Science in Robotics* | GPA: 3.86/4

Sep 2025 - May 2027

Savitribai Phule Pune University | *Bachelor of Engineering in Mechanical* | GPA: 3.5/4

Sep 2019 - Jul 2023

SKILLS

Languages & Middleware : Python, C++, Embedded C, SQL, MATLAB, ROS/ROS2

Robotics : RTAB, ORB-SLAM3, PCL, OpenCV, PyTorch, TensorFlow, YOLO, AprilTag, Nav2, AMCL, DWA, RViz

Software : Gazebo, MuJoCo, SolidWorks, Simulink

Hardware : Modbus, TCP/IP, I2C, LiDAR, RGB-D Camera, IMU/Gyroscope, Motor control, Sensor Calibration

Tools : Linux, Git, scikit-learn, NumPy, SciPy, Pandas, Matplotlib

Certification : OpenCV Bootcamp, Autonomous Robots (NVIDIA), Foundations of IoT Systems & Industrial Automation

EXPERIENCE

Innomotics India Pvt. Ltd. (formerly Siemens Large Drive) | *Manufacturing Professional*

Jul 2023 - Jul 2025

- Supervised end-to-end production of **3.3KV-11KV** medium-voltage motors, coordinating supplier timelines, tracking lead times, and generating production schedules to scale monthly output from **10** to **25** units
- Analyzed **2** years of **1000+** order records using **Advanced Excel** to classify A/B/C material demand, implementing **Kanban/JIT** to improve monthly output by **100%**
- Automated **MIS reporting** by centralizing cross-department updates in one dashboard, reducing report cycle time by **40%**
- Redesigned kitting trolley layout using **shadow boards**, barcode scanning, and material kitting tracker, reducing manual search time by **25%** and enabling real-time material stage visibility across assembly stations
- Tested and validated **3.3KV-11KV** VFDs with **hardware configuration**, ensuring product reliability across assembly

Wissenquelle - OneStop Ai | *Robotics Project Intern*

Feb 2022 - Apr 2022

- Derived **DH parameters** and computed **forward kinematics** for 6-DOF PUMA 560, 3-DOF planar manipulator, and 4-DOF SCARA in **RoboAnalyzer**, determining end-effector poses across multiple joint configurations
- Validated kinematic models against simulation, verifying spatial consistency of joint-space to task-space mappings

PROJECTS

GPS-Denied Drone SLAM for Underground Mine Mapping | *ROS2, LIO-SAM, RTAB-Map, PCL, Gazebo, Nav2*

[GitHub](#)

- Integrated LIO-SAM + IMU pipeline to achieve globally consistent 3D reconstruction across cave meshes
- Constructed RTAB-Map loop closure pipeline across **10,578 nodes**, generating a complete 3D point cloud reconstruction of the tunnel at **0.01 m** voxel resolution
- Deployed frontier-based exploration via Nav2 + OctoMap, achieving 100% waypoint coverage across cave topology
- Trained k-NN/SVM NIR spectral classifier identifying 5 mineral types at **81-93% confidence**, georeferencing **2,617 detection points** as 3D markers on the live OctoMap

Autonomous Visual Navigation in Dynamic Environments | *ROS2, MuJoCo 3.5, Nav2, OpenCV, Python, AMCL, DWA*

[GitHub](#)

- Assembled a **differential-drive robot** with camera and Lidar, for wayfinding in a **10×10m arena** with dynamic obstacles
- Established **ROS2 Humble** bridge publishing sensor data, integrating **Nav2's** adaptive monte carlo for path planning
- Modeled **3 dynamic obstacles** with Brownian motion, with deterministic patterns to simulate uncertain real-world objects
- Implemented HSV color thresholding for goal detection, to maintain stable target detection under Gaussian lighting noise across all 5 tested noise levels ($\sigma = 0.0-0.8$)

Real-Time Pedestrian Trajectory Prediction | *RTMPose, SMPL, Social-LSTM, CVAE, PyTorch*

[GitHub](#)

- Engineered dual-stream architecture combining **Social-LSTM** with Transformer pose encoder, achieving **35-46%** error reduction on JTA dataset and **0.67m minADE** on ETH/UCY benchmark
- Deployed **RTMPose + SMPL** for 3D pose estimation at **30 FPS**, extracting 17 keypoints to predict pedestrian intent **0.8-1.0s earlier**
- Developed **CVAE decoder** generating K=20 multimodal trajectories over a 4.8s horizon

3D Scene Reconstruction via Structure-from-Motion | *SIFT, Grabcut, RANSAC, EPnP, Bundle Adjustment, OpenCV*

[GitHub](#)

- Developed a SfM pipeline applying SIFT with GrabCut, extracting **1199 keypoints per image** across a 62-image dataset
- Implemented **Lowe's ratio test** and RANSAC estimation to reject outliers, achieving $\geq 70\%$ inlier ratios
- Reconstructed camera trajectory via Essential Matrix decomposition, triangulating a **12,968-point** sparse point cloud
- Achieved **0.83 px mean reprojection error** by sparse bundle adjustment with COO Jacobian, producing an **8,672-point** point cloud in Meshlab

RESEARCH

Pose-Guided Trajectory Prediction (PGTP-Net) | *Ongoing*

[Draft](#)

- Extending a dual-stream **Social-LSTM + transformer pose** architecture toward full **3D pose-conditioned** pedestrian forecasting, integrating **SMPL**, **HMR2.0**, and **WHAM** for monocular 3D pose recovery